

# Counterexample: Semi $w^*$ -Denting Points Need Not Be Extreme

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## Source Question

The source paper is Sudeshna Basu and Susmita Seal, *Ball separation characterization of ball dentability and related properties*, arXiv:2307.03621, version 4 [1].

Definition 4.1 defines semi denting points and their weak-star analogues. Remark 4.3 observes that if  $x_0^* \in S_{X^*}$  is a semi  $w^*$ -denting point of  $B_{X^*}$ , then  $x_0^*$  is a norm limit of extreme points of  $B_{X^*}$ . The same remark says that it is not known whether every semi  $w^*$ -denting point must be extreme. In the parsed source this is `data/parsed/arxiv_sources/2307.03621/Ball_sep_arxiv.tex`, lines 818–835, and in the PDF it is page 13.

- (i) [12] *semi denting point* of  $B_X$  if for every  $\varepsilon > 0$ , there exists slice  $S$  of  $B_X$  such that  $S \subset B(x, \varepsilon)$ .
- (ii) *semi Point of Continuity* (semi PC) of  $B_X$  if for every  $\varepsilon > 0$ , there exists a weakly open set  $V$  in  $B_X$  such that  $V \subset B(x, \varepsilon)$ .
- (iii) *semi Small Combination of Slice* (semi SCS) point of  $B_X$  if for every  $\varepsilon > 0$ , there exists a convex combination of slices  $\sum_{i=1}^n \lambda_i S_i$  in  $B_X$  such that  $\sum_{i=1}^n \lambda_i S_i \subset B(x, \varepsilon)$ .

Analogously, we can define semi  $w^*$ -denting point, semi  $w^*$ -PC and semi  $w^*$ -SCS of  $B_{X^*}$ .

**Remark 4.2.** It is clear that the existence of a semi denting (respectively, semi PC) point in  $B_X$  implies that  $X$  has the BDP (respectively, BHP). However, it is not known whether the converse holds in general. It is true that  $X$  has *BSCSP* if and only if  $0$  is a SCS point in  $B_X$ . Indeed, if  $X$  has *BSCSP*, then for  $\varepsilon > 0$  there exists a convex combination of slices  $T$  of  $B_X$  with diameter less than  $\varepsilon$ . The convex combination of slices  $T' = \frac{1}{2}T + \frac{1}{2}(-T)$  of  $B_X$  then contains  $0$  and has diameter less than  $\varepsilon$ . It remains unknown whether *BSCSP* ensures the existence of any nonzero SCS point or nonzero semi SCS point in  $B_X$ .

**Remark 4.3.** If a point  $x_0^* \in S_{X^*}$  is a semi  $w^*$ -denting point of  $B_{X^*}$ , then there exists a sequence  $(x_n^*)_n$  of extreme points in  $S_{X^*}$  that converges to  $x_0^*$  in norm. Indeed, let  $x_0^* \in S_{X^*}$  be a semi  $w^*$ -denting point of  $B_{X^*}$ . Then there exists a sequence  $(S_n)_n$  of  $w^*$ -slices of  $B_{X^*}$  such that  $S_n \subset B(x_0^*, \frac{1}{n})$  for all  $n \in \mathbb{N}$ . By the Krein-Milman Property (KMP) and the extremal structure of slices (that is, if  $S$  is a slice of  $B_{X^*}$  and  $\sum_{i=1}^l \lambda_i a_i \in S$  with  $0 < \lambda_i < 1$  and  $\sum_{i=1}^l \lambda_i = 1$ , then  $a_i \in S$  for some  $i$ ), each  $S_n$  contains an extreme point  $x_n^*$  (say) of  $B_{X^*}$ . Consequently,  $(x_n^*)_n$  converges to  $x_0^*$  in norm. However, unlike the case of  $w^*$ -denting points, it is not known whether every semi  $w^*$ -denting point must be an extreme point.

**Theorem 4.4.** [16, Theorem 4.1] *Let  $X$  be a Banach space and  $x_0^* \in S_{X^*}$ . Then  $x_0^*$  is semi  $w^*$ -denting point of  $B_{X^*}$  if and only if for every bounded set  $C \subset X$  with  $\inf x_0^*(C) > 0$ , there exists a closed ball  $B$  in  $X$  such that  $C \subset B$  and  $0 \notin B$ .*

## Idea of the Proof

The definition of a semi denting point is deliberately one-sided: for each  $\varepsilon > 0$  there must be a small slice contained in  $B(x, \varepsilon)$ , but the slice need not contain  $x$ . Therefore every norm limit of denting points is automatically semi denting. To disprove the extreme-point conclusion, it is enough to

build a finite-dimensional dual ball with a non-extreme boundary point that is a norm limit of exposed points. Finite dimensionality then makes the weak-star and norm topologies coincide.

## Construction

Work in  $\mathbb{R}^4$  with coordinates  $(s, u, v, t)$ . For  $0 \leq r \leq 1$ , put

$$c(r) = (0, r, r^2, 1), \quad a = (-1, 0, 0, 1), \quad b = (1, 0, 0, 1),$$

and define

$$F = \text{co}(\{a, b\} \cup \{c(r) : 0 \leq r \leq 1\}), \quad K = \text{co}(F \cup (-F)).$$

The set  $K$  is compact, convex, centrally symmetric, and full dimensional. Hence  $K$  is the closed unit ball of a four-dimensional real Banach space  $Y$ . Let

$$p = c(0) = (0, 0, 0, 1).$$

Since  $p = (a+b)/2$  and  $a \neq b$ ,  $p$  is not an extreme point of  $K$ . Also  $p \in S_Y$ , because the coordinate functional  $t$  satisfies  $\sup_K t = 1$  and  $t(p) = 1$ .

**Lemma.** *For every  $r_0 \in (0, 1]$ , the point  $c(r_0)$  is an exposed point of  $K$ .*

*Proof.* Fix  $r_0 \in (0, 1]$  and set

$$g_{r_0}(s, u, v, t) = 2r_0u - v.$$

On the curve  $c(r)$ ,

$$g_{r_0}(c(r)) = 2r_0r - r^2 = r_0^2 - (r - r_0)^2 \leq r_0^2,$$

with equality only when  $r = r_0$ . On the segment  $[a, b]$ , the value of  $g_{r_0}$  is  $0 < r_0^2$ . By linearity,  $g_{r_0}$  has the unique maximum  $r_0^2$  on  $F$  at  $c(r_0)$ .

Let  $M = \sup\{|g_{r_0}(y)| : y \in F\}$ , and choose  $\lambda > 0$  so small that  $-1 + \lambda M < 1$ . Consider

$$\Phi(y) = t(y) + \lambda g_{r_0}(y).$$

On  $F$ , where  $t = 1$ , the unique maximum of  $\Phi$  occurs at  $c(r_0)$ . On  $-F$ , where  $t = -1$ , we have  $\Phi(y) \leq -1 + \lambda M < 1$ , while  $\Phi(c(r_0)) = 1 + \lambda r_0^2 > 1$ . Thus  $c(r_0)$  is the unique maximizer of  $\Phi$  on  $K$ , so it is exposed.  $\square$

## Counterexample

**Counterexample.** *There exists a Banach space  $X$  and a point  $p \in S_{X^*}$  such that  $p$  is a semi  $w^*$ -denting point of  $B_{X^*}$ , but  $p$  is not an extreme point of  $B_{X^*}$ .*

*Proof.* Let  $Y$  be the four-dimensional Banach space whose unit ball is the set  $K$  constructed above, and let  $X = Y^*$ . Since  $Y$  is finite dimensional, we identify  $X^*$  canonically with  $Y$ , and the weak-star topology on  $X^* = Y$  agrees with the norm topology.

We already observed that  $p = c(0) \in S_Y$  is not extreme in  $K = B_Y$ . It remains to show that  $p$  is semi denting, hence semi  $w^*$ -denting under the identification  $X^* = Y$ .

For each  $n \geq 1$ , the point  $c(1/n)$  is exposed by the lemma. In a compact finite-dimensional convex set, every exposed point is a denting point: if a linear functional has a unique maximum at  $y$ , compactness implies that the slices

$$\{z \in K : \phi(z) > \phi(y) - \alpha\}$$

shrink in norm to  $y$  as  $\alpha \downarrow 0$ . Hence each  $c(1/n)$  is a denting point of  $K$ .

The sequence  $c(1/n)$  converges to  $p$  in the Euclidean topology, and therefore in the norm of  $Y$ . Given  $\varepsilon > 0$ , choose  $n$  such that  $\|c(1/n) - p\|_Y < \varepsilon/2$ . Since  $c(1/n)$  is denting, choose a slice  $S$  of  $K$  containing  $c(1/n)$  with diameter less than  $\varepsilon/2$ . Then

$$S \subset B_Y(c(1/n), \varepsilon/2) \subset B_Y(p, \varepsilon).$$

This is exactly the semi denting condition for  $p$ . Since weak-star and norm topologies coincide in finite dimension,  $p$  is a semi  $w^*$ -denting point of  $B_{X^*}$ . But  $p$  is not extreme, as  $p = (a + b)/2$  with  $a \neq b$ .  $\square$

## Verification Notes

The proof is finite-dimensional and does not rely on unproved external theorems beyond standard compact-convex facts: exposed points of compact finite-dimensional convex bodies are denting points, and all Hausdorff vector topologies on a finite-dimensional space coincide. The construction is also stable under replacing the continuum  $\{c(r) : 0 \leq r \leq 1\}$  by a sufficiently rich compact convex curve with exposed points accumulating at  $c(0)$ .

## Novelty Search Bounds

Before promotion, the run registry, solution index, attempt index, and proof-gap index were searched for arXiv:2307.03621, ball separation, dentability, semi  $w^*$ -denting, extreme point, SCS, and related terms. No existing packet for this question was found. A web search on June 16, 2026 for close phrases including “semi  $w^*$  denting” with “extreme point”, “semi weak\* denting” with “extreme point”, and “semi denting point” with “extreme point” did not reveal an exact prior answer. This is a bounded novelty check, not a bibliographic proof of novelty.

## Human Review Recommendation

The main point for review is the exposure argument for  $c(r_0)$  and the interpretation of semi denting as allowing the small slice to lie near  $p$  without containing  $p$ . If both are accepted, the counterexample directly answers the question in Remark 4.3 negatively.

## References

- [1] S. Basu and S. Seal, *Ball separation characterization of ball dentability and related properties*, arXiv:2307.03621, version 4, 2025.